

Basic Applications of Operational Amplifiers



Fall – 2024/2025

ELECTRICAL CIRCUITS

Lab 4 Report

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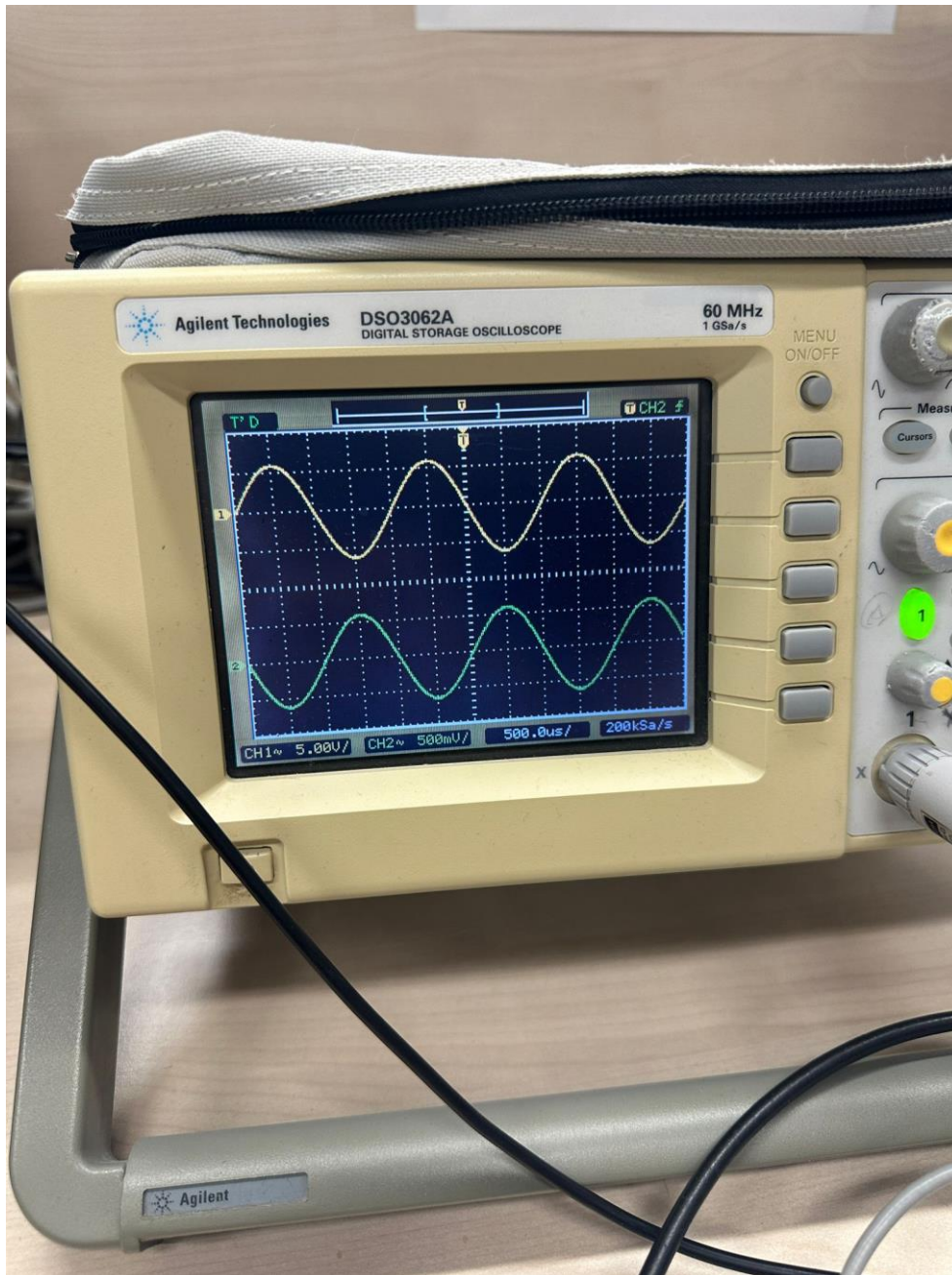
Objective:

Participants in this experiment will learn the fundamentals of using operational amplifiers.

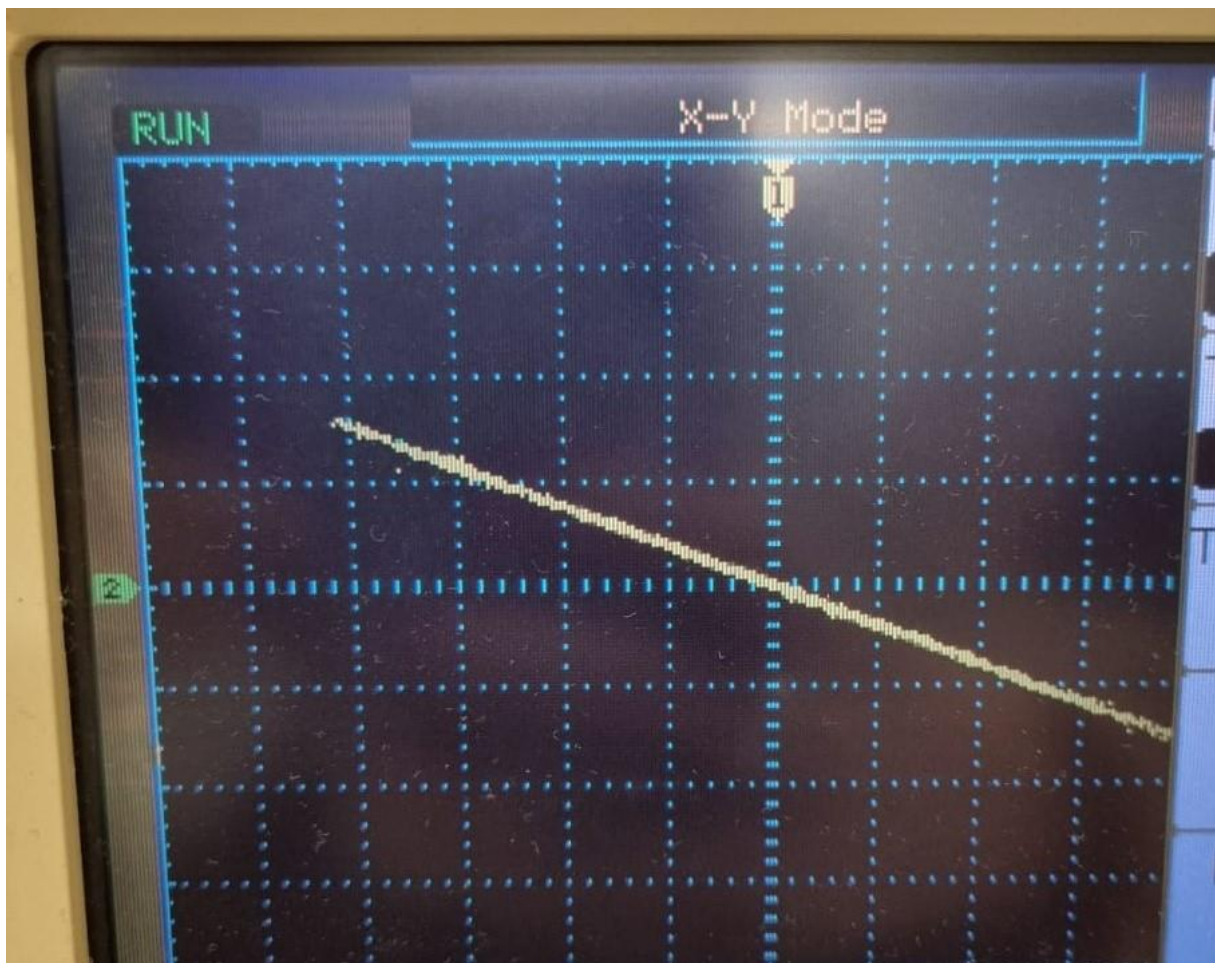
Additionally, various resistor configurations with node voltage analysis will be used to cover the characteristics of circuits with op-amps. Furthermore, the fundamental

We'll go over the requirements and limitations of op-amps, how to use an oscilloscope and function generator to analyze AC signals, and feedback methods to produce op-amps with the desired performance.

Task-1

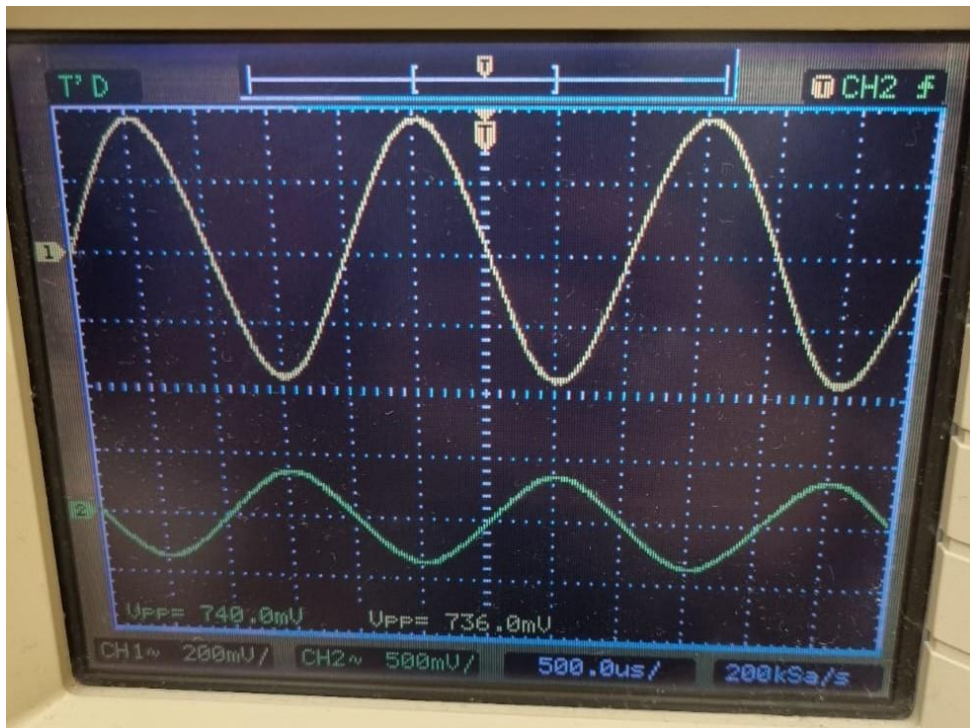


Observation of $v_{in}(t)$ and $v_o(t)$ on the oscilloscope. In this image, the yellow waveform (CH1) represents the input voltage V_{in} , while the green waveform (CH2) represents the output voltage V_o .

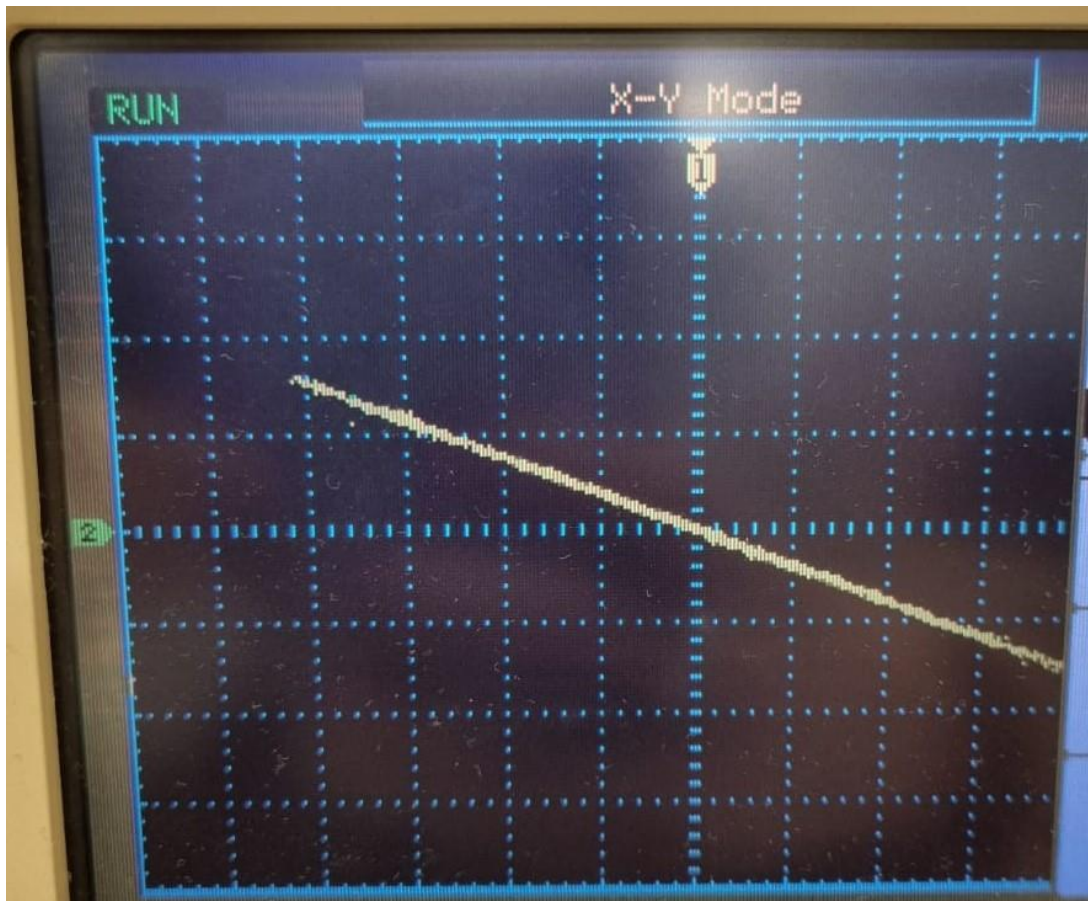


Observation of XY mode. In X-Y mode, the linear relationship between the input voltage (X-axis) and the output voltage (Y-axis) is displayed.

Task-2



Observation of $v_{in}(t)$ and $v_o(t)$ on the oscilloscope. In this image, the yellow waveform (CH1) represents the input voltage V_{in} , while the green waveform (CH2) represents the output voltage V_o .



Observation of XY mode. In X-Y mode, the linear relationship between the input voltage (X-axis) and the output voltage (Y-axis) is displayed.

Conclusion:

We built inverting and summing circuits on breadboard and then we connected to oscilloscope in order to see the V_{in} and V_{out} values as waveform and also we did the observation on XY mode as well.